

Azure Data Pipeline using Azure Data Factory

Objective

The objective of this project is to understand Azure cloud services and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory. The project demonstrates how data can be stored, validated, and copied from one Blob container to another using Azure Data Factory activities.

Introduction

Microsoft Azure is a cloud computing platform that provides various services such as storage, networking, and data integration. Azure Data Factory is a cloud-based ETL service used for creating data pipelines.

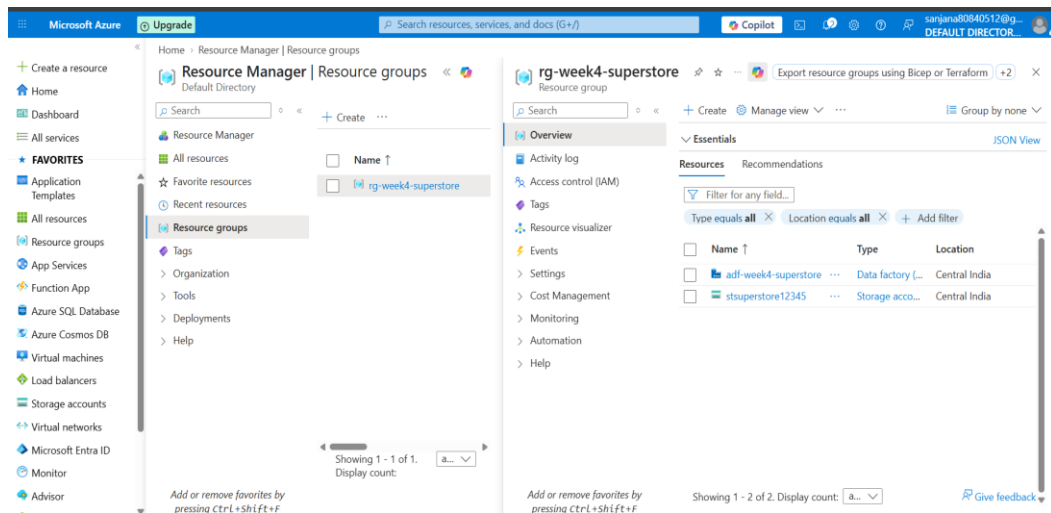
In this project, a CSV file is stored in Azure Blob Storage. Azure Data Factory validates the file using Get Metadata activity and copies it to another Blob container using Copy Data activity.

Azure Services Used

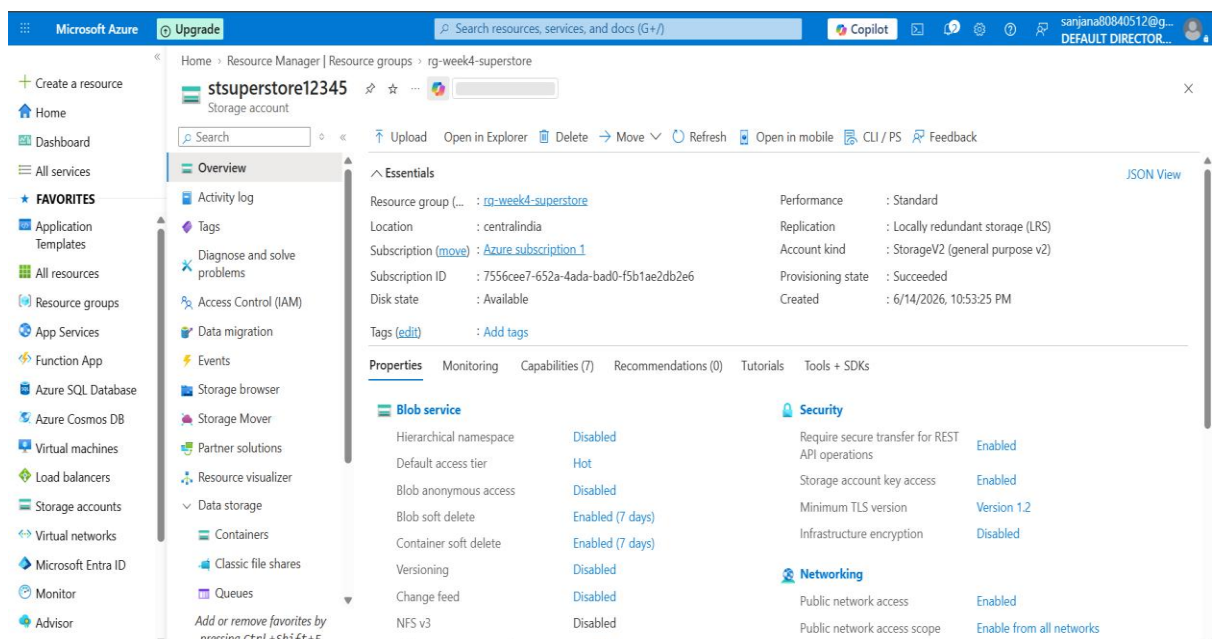
- Azure Resource Group
- Azure Storage Account
- Blob Storage Containers
- Azure Data Factory
- Linked Service
- Source Dataset
- Destination Dataset
- Get Metadata Activity
- Copy Data Activity
- IAM (Identity and Access Management)

Project Implementation

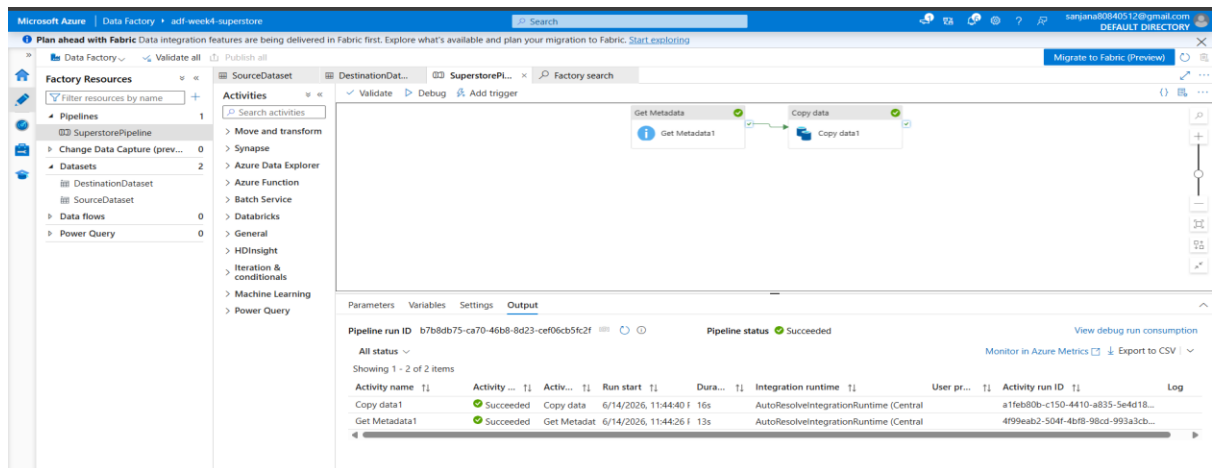
- Created a Resource Group in Azure Portal.



- Created a Storage Account.
- Created two Blob Containers
 1. raw-data
 2. processed-data
- Uploaded the Superstore CSV dataset into the raw-data container.
- Created an Azure Data Factory instance.
- Configured Linked Service to connect Azure Data Factory with Blob Storage.

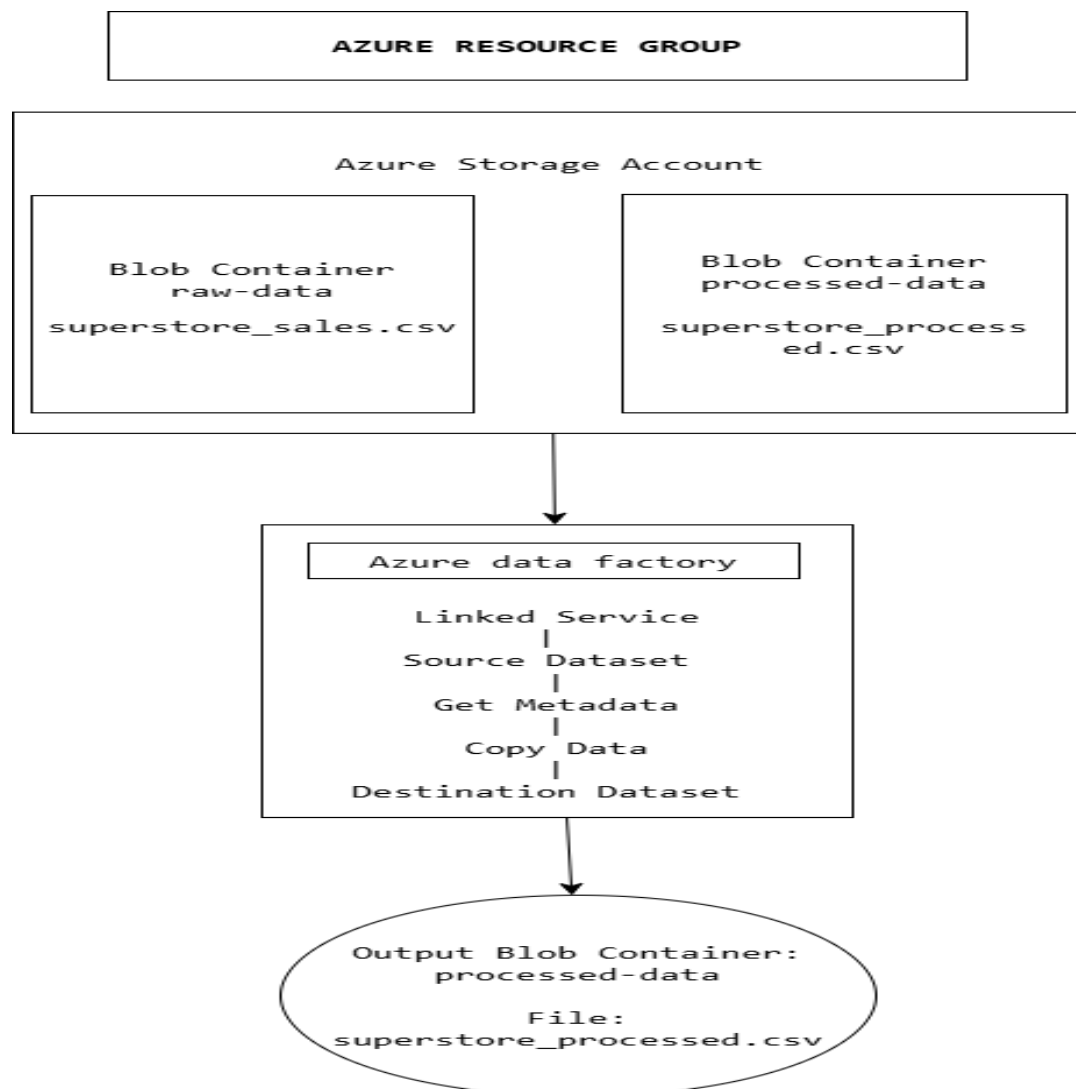


- Created Source Dataset and Destination Dataset.
- Added Get Metadata activity to validate the source file.
- Added Copy Data activity to copy the CSV file from raw-data container to processed-data container.
- Executed the pipeline using Debug and verified successful execution.



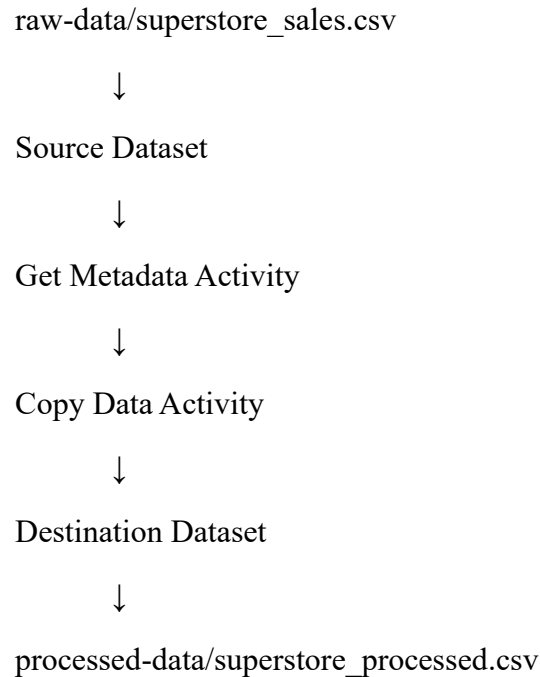
- Assigned IAM roles to provide access between Azure Data Factory and Azure Storage.

Architecture



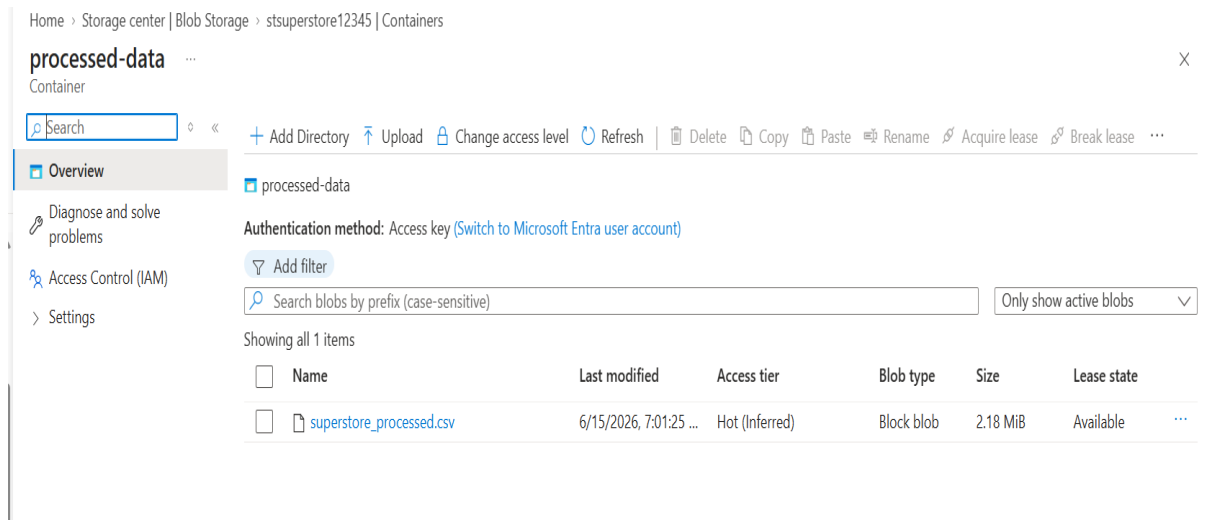
The architecture consists of Azure Blob Storage as the source and destination, connected through Azure Data Factory. The pipeline first validates the source file using Get Metadata activity and then copies the data to the destination container using Copy Data activity.

Pipeline Flow



Results

- Resource Group created successfully.
- Storage Account created successfully.
- Blob containers created successfully.
- CSV file uploaded successfully.
- Linked Service configured successfully.
- Source and Destination datasets created successfully.
- Get Metadata activity validated the source file.
- Copy Data activity copied the file successfully.
- Pipeline execution completed successfully.
- Output file generated in the processed-data container.
- IAM roles assigned successfully.



Conclusion

This project successfully demonstrated the implementation of an end-to-end data pipeline using Azure Data Factory. The pipeline validated the source CSV file and copied it from the raw-data container to the processed-data container. The project provided practical understanding of Azure Storage, Azure Data Factory, datasets, linked services, pipeline execution, and IAM role management.